

M2 COMMUNITY NODE

Rack Wiring & Cable Routing Diagram

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Power chain · Data routing · RF paths · EMI separation zones

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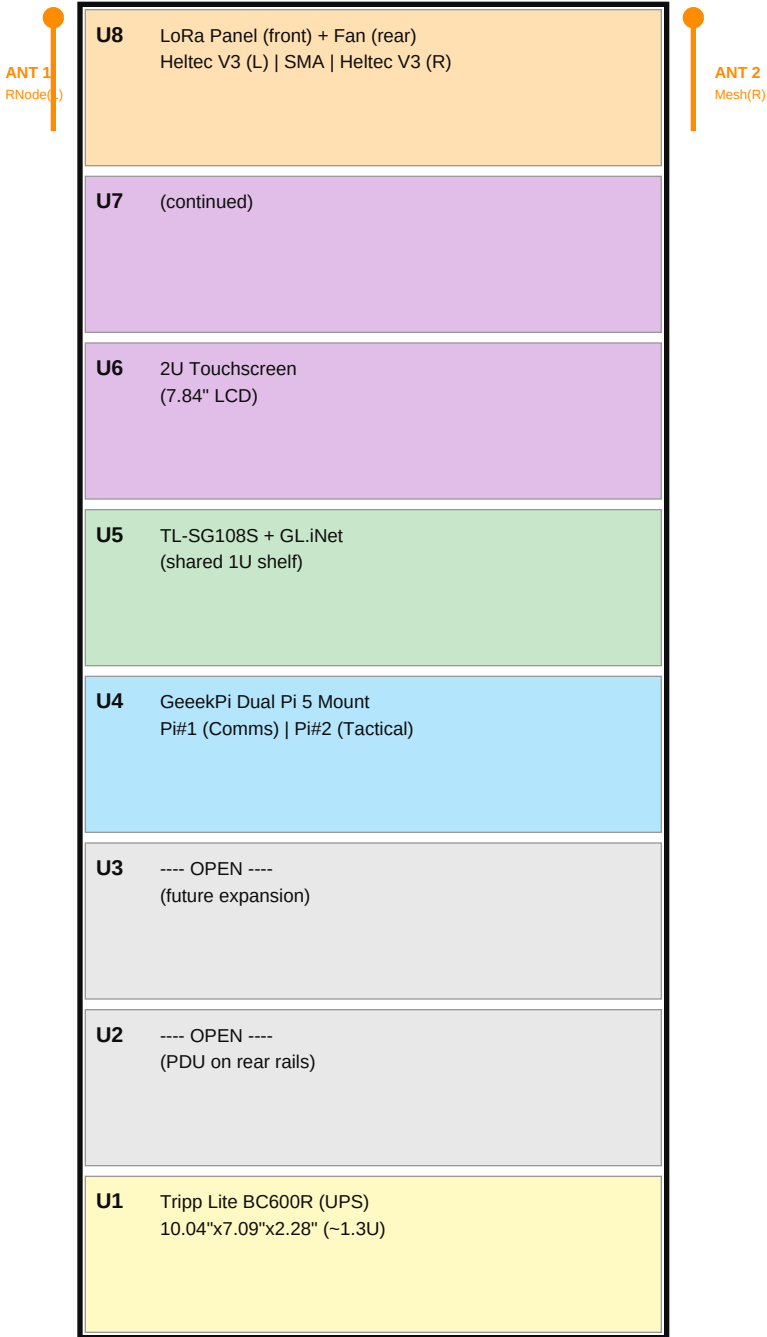
M2 COMMUNITY NODE — RACK WIRING DIAGRAM

GeeekPi RackMate T18U | Front View + Side Cross-Section | Cable Routing

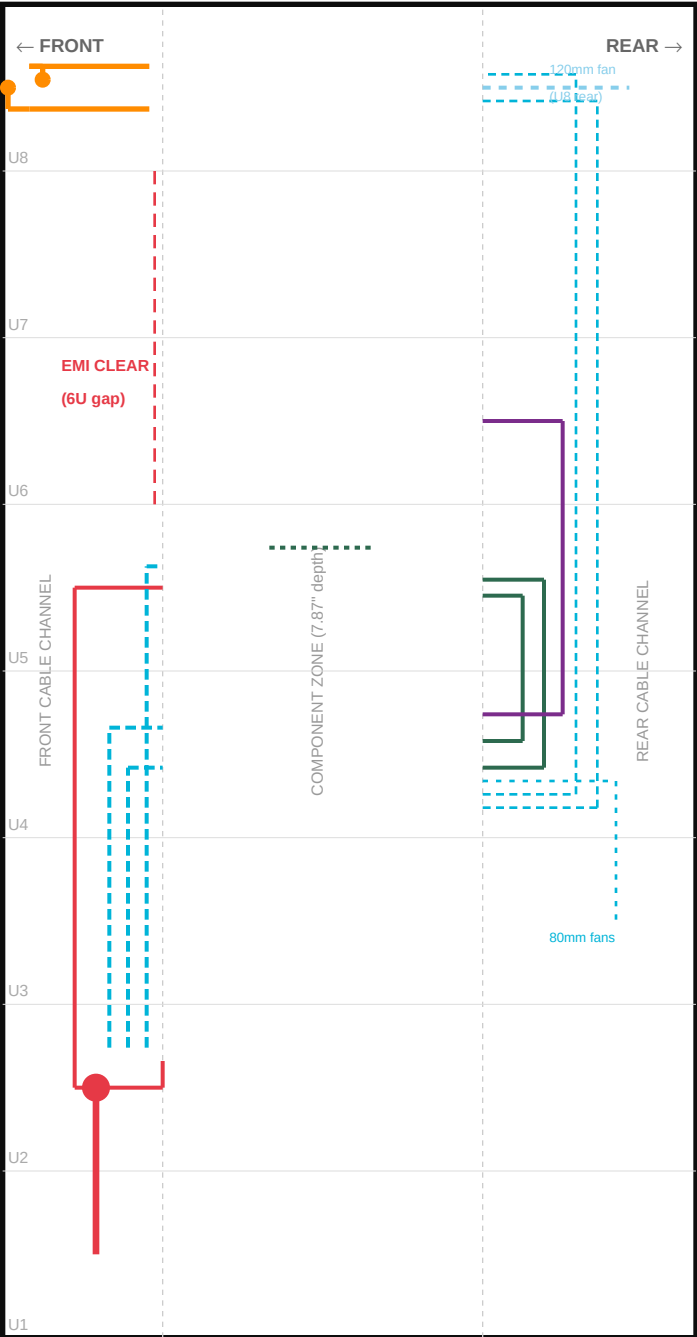
CABLE LEGEND:

- AC Power (UPS U1 -> PDU U2 rear rails; PDU -> Switch 12V barrel U5)
- USB Power/Data (Anker -> Pi/GL.iNet USB-C, Pi -> LoRa 3ft USB-C, Hub -> Fans)
- Ethernet (Switch U5 <-> Pi#1, Pi#2 U4, GL.iNet U5 -- 6" patches)
- RF / Antenna (U.FL pigtails -> SMA bulkheads -> 915 MHz antennas at U8 top)
- HDMI (Pi#1 micro HDMI U4 -> Touchscreen U7, 2ft cable)

FRONT VIEW (Top-Down)



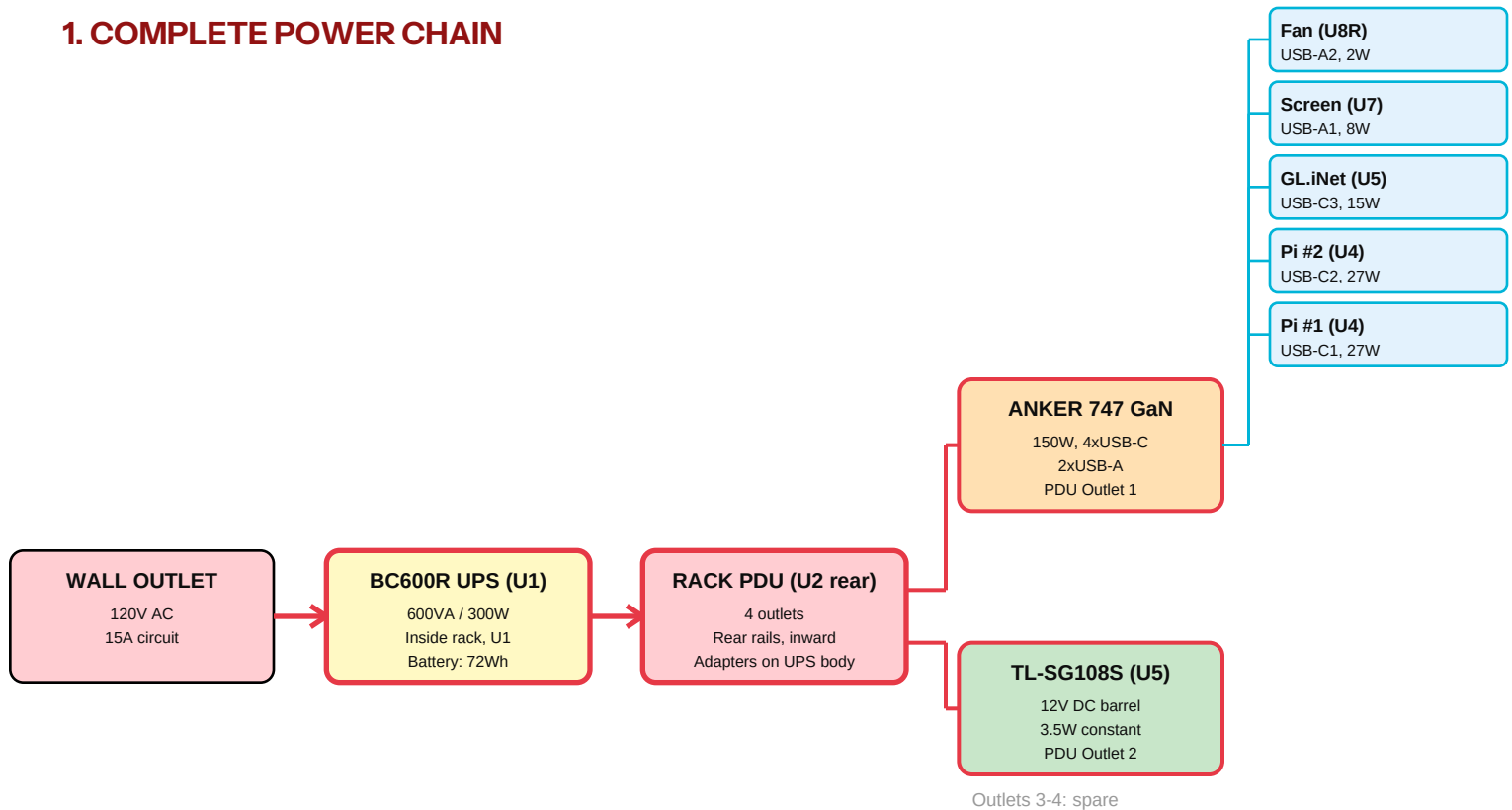
SIDE CROSS-SECTION (Cable Routing)



POWER CHAIN & EMI SEPARATION

Wall -> UPS (U1) -> PDU (U2 rear) -> Devices | Cable Routing Rules | RF Protection Zones

1. COMPLETE POWER CHAIN



2. EMI SEPARATION ANALYSIS

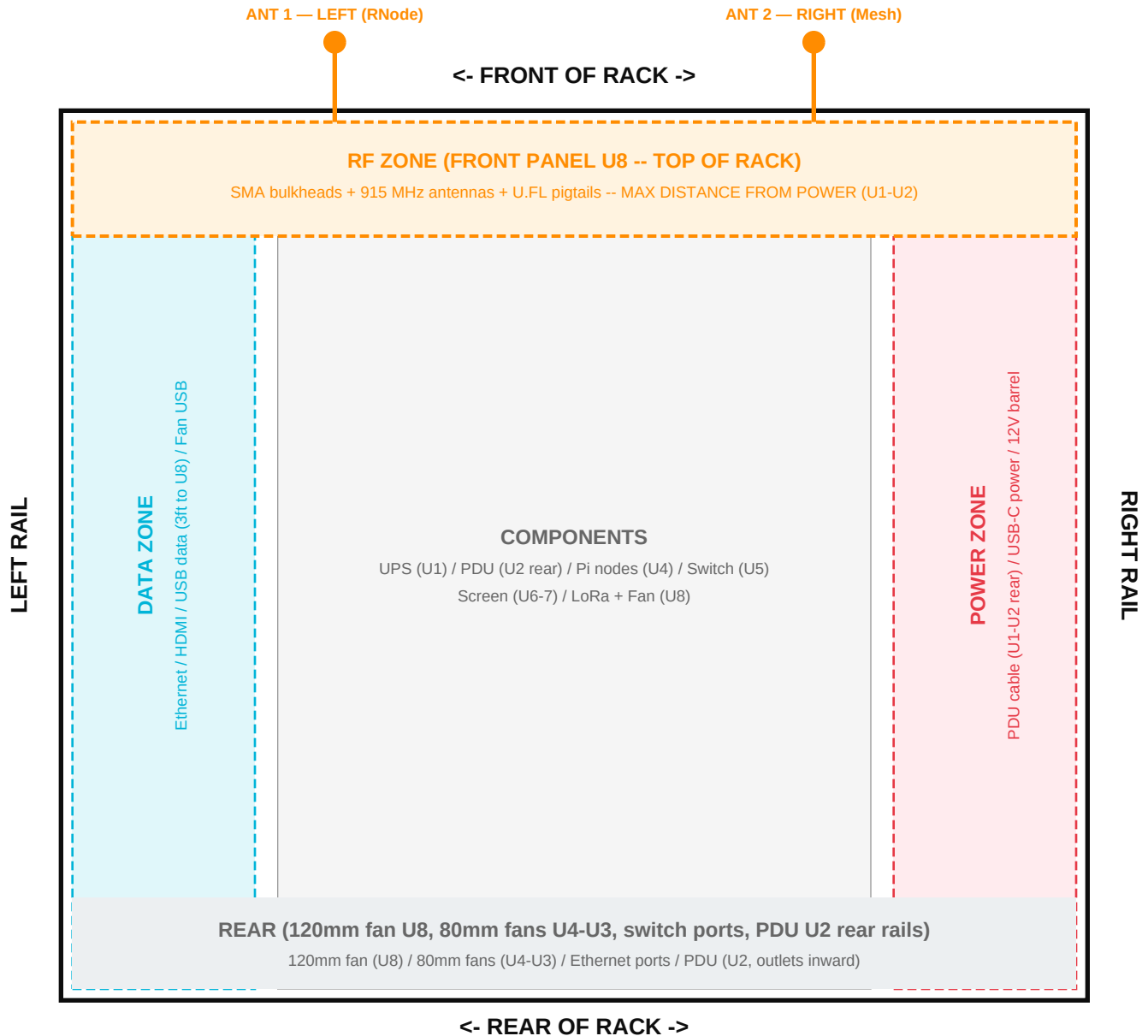
- ✓ GOOD: LoRa at U8 (top), PDU at U2 rear -- 6U (10.5") vertical separation
- ✓ GOOD: LoRa at U8 (top), UPS at U1 -- 7U (12.25") maximum separation
- ✓ GOOD: Power cables concentrated at U1-U2 bottom; RF at U8 top -- opposite ends
- ✓ GOOD: AC power cable is shielded (PDU 6ft cord has integral ground)
- ✓ GOOD: U.FL pigtails are < 6" long -- minimal exposure to radiated EMI
- ✓ GOOD: 915 MHz LoRa is spread-spectrum (chirp) with ~19 dB noise immunity
- ✓ GOOD: USB data cables (Pi U4 -> LoRa U8, 3ft) route via LEFT rail -- away from power
- ! CAUTION: 3ft USB-C data cables traverse 4U on left rail
- > MITIGATION: Route USB data cables tightly along left rail; cross power paths at 90 deg only

3. CABLE ROUTING RULES

- LEFT RAIL (rear): Ethernet patches (6"), HDMI (2ft, Pi U4 -> Screen U7), fan USB cables
- LEFT CHANNEL: USB-C data cables (Pi#2 U4 -> LoRa boards at U8, 3ft)
- RIGHT CHANNEL: USB-C power cables (Anker -> Pi#1 U4, Pi#2 U4, GL.iNet U5, Screen U7)
- RIGHT RAIL (rear): AC power (PDU 6ft cord U1->U2 rear rails, ultra-short), 12V barrel adapter cable
- FRONT PANEL (U8): RF only -- SMA bulkheads + antennas, OLED windows (TOP of rack)
- RULE: Never bundle AC power and RF cables together. Cross at 90 deg only.**

TOP-DOWN VIEW — CABLE SEPARATION ZONES

Looking down into the rack from U8 | Data LEFT, Power RIGHT, RF FRONT (U8 top)



KEY ROUTING PRINCIPLE:

Data cables (Ethernet, HDMI, USB data) run on the LEFT rail/channel only.

Power cables (AC + USB-C power) run on the RIGHT rail/channel only.

RF paths (U.FL pigtails, SMA bulkheads, antennas) stay at the FRONT panel, U8 (TOP of rack).

Power at U1-U2, RF at U8 = 6U (10.5") separation. Excellent EMI isolation.